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(54) **AIRCRAFT DOOR AND PRODUCTION METHOD**

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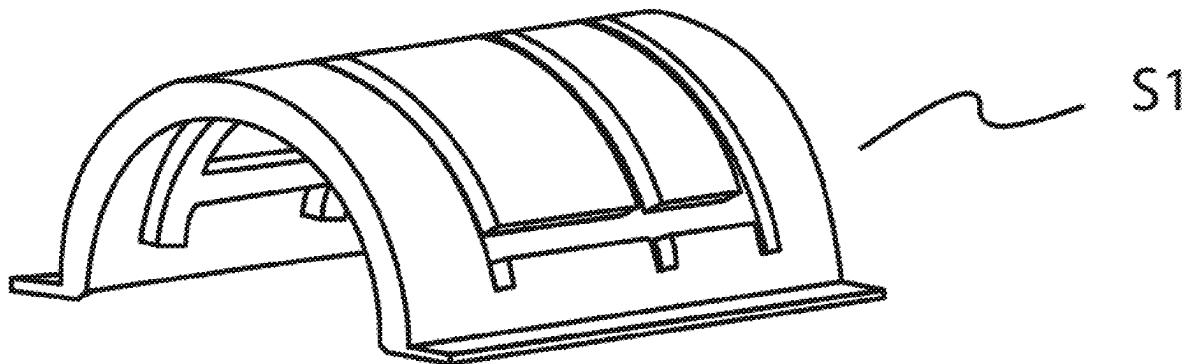
(2013.01); **B32B 2605/18** (2013.01); **B64C**

2001/0072 (2013.01); **B64D 29/08** (2013.01)

(57)

ABSTRACT

A door for aircraft, configured to be moved from a closed position of an outer shell of a nacelle to an open position to allow an inspection of an internal compartment of the nacelle. The door includes a first multilayer portion, including an inner layer, an outer layer and an intermediate layer, the intermediate layer is made of composite material having an inorganic-based matrix and a carbon-based filler, the first portion shaped to have at least one substantially oblong stiffening protuberance; a second layered portion having an aerodynamic profile and shaped to be coupled to the first multilayer portion, the overall configuration of the door such that in an assembled configuration, and in a coupling between the first portion and the second portion, the substantially oblong stiffening protuberance faces the internal compartment of the nacelle.



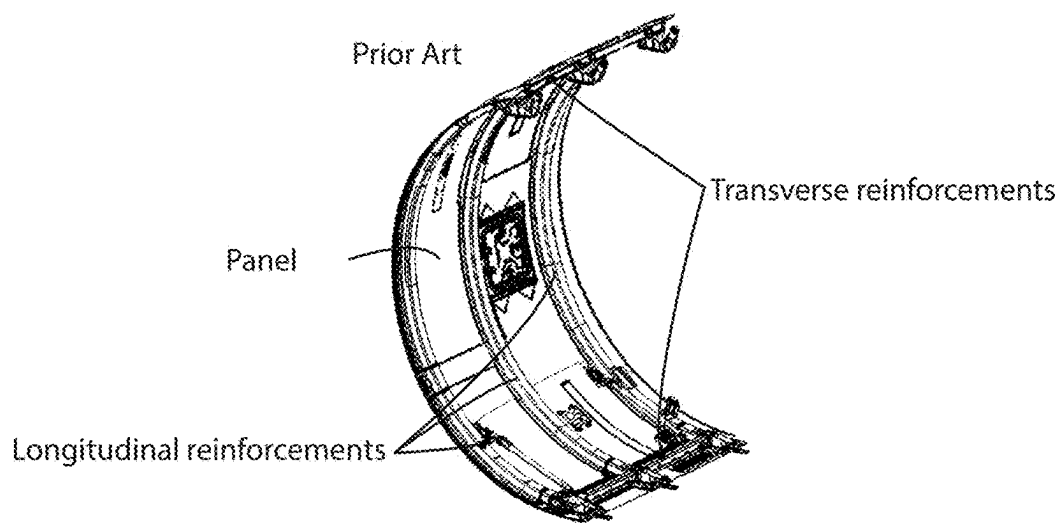


Fig. 1

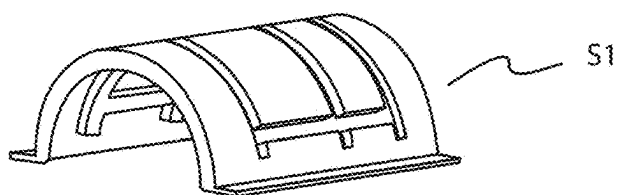


Fig. 2a

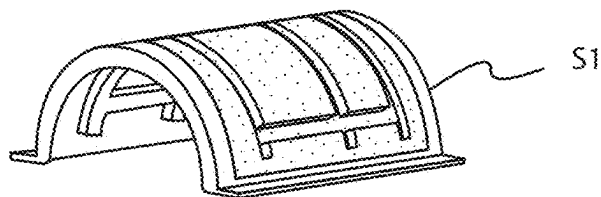


Fig. 2b

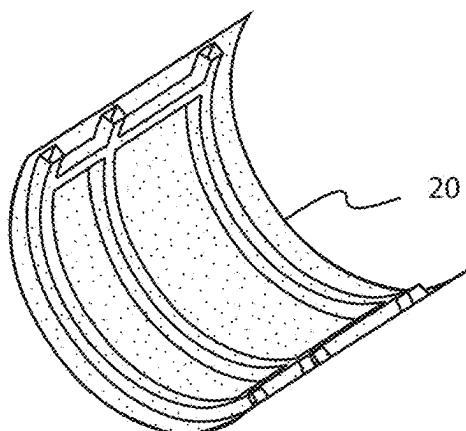


Fig. 2c

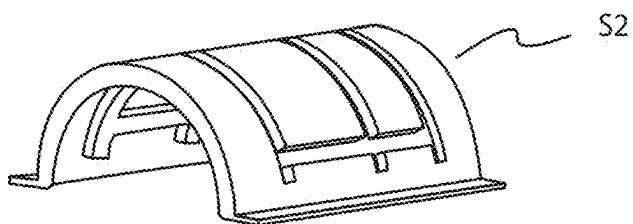


Fig. 3a

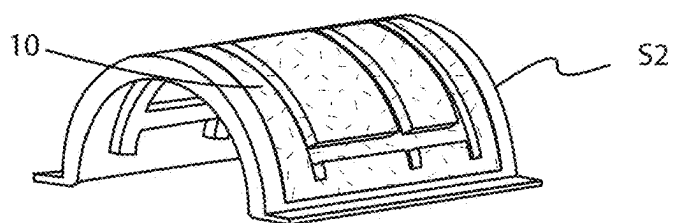


Fig. 3b

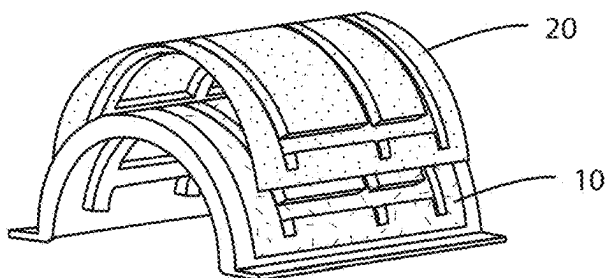


Fig. 3c

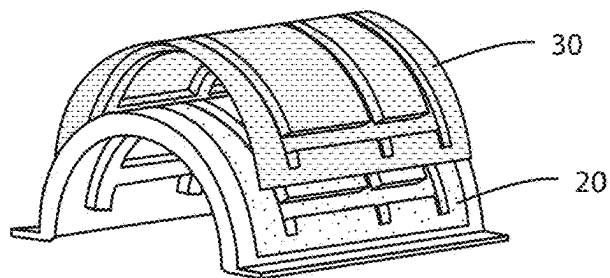


Fig. 3d

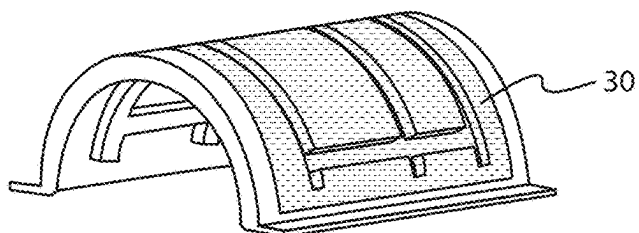


Fig. 3e

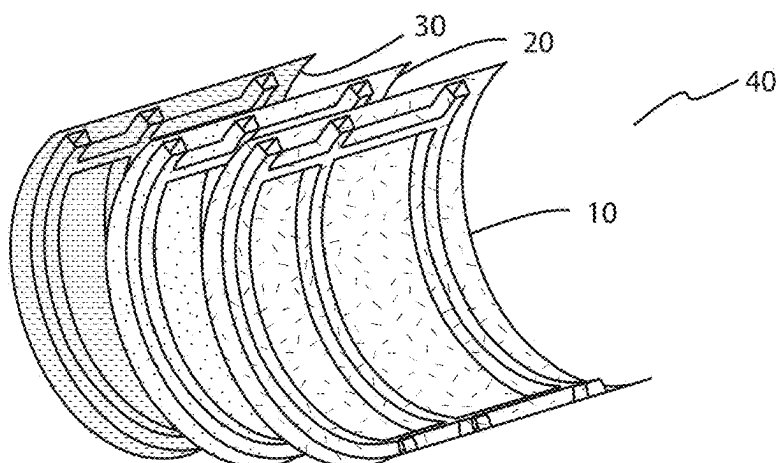


Fig. 4

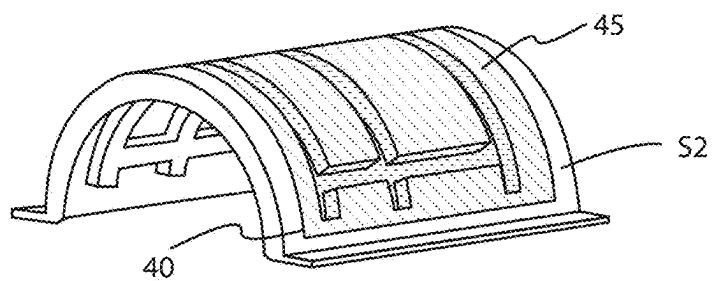


Fig. 5

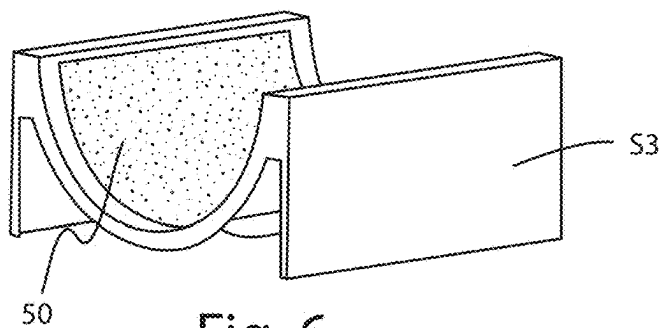


Fig. 6

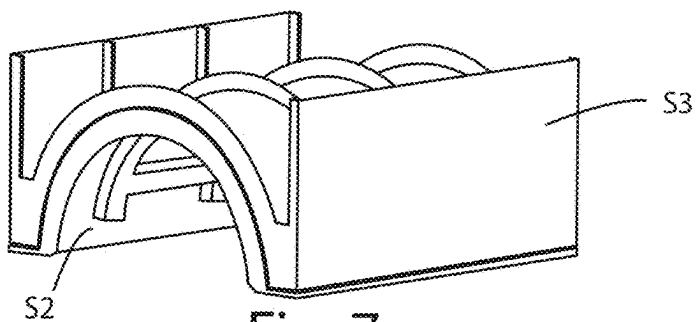


Fig. 7

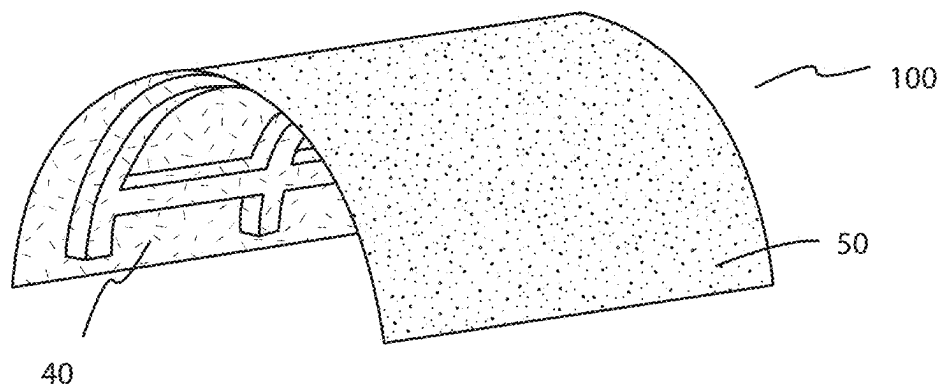


Fig. 8

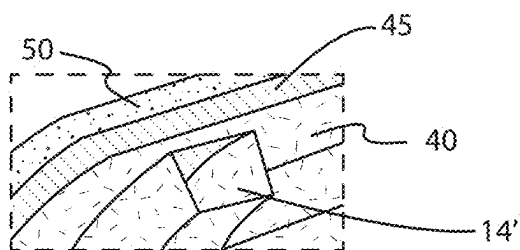


Fig. 9b

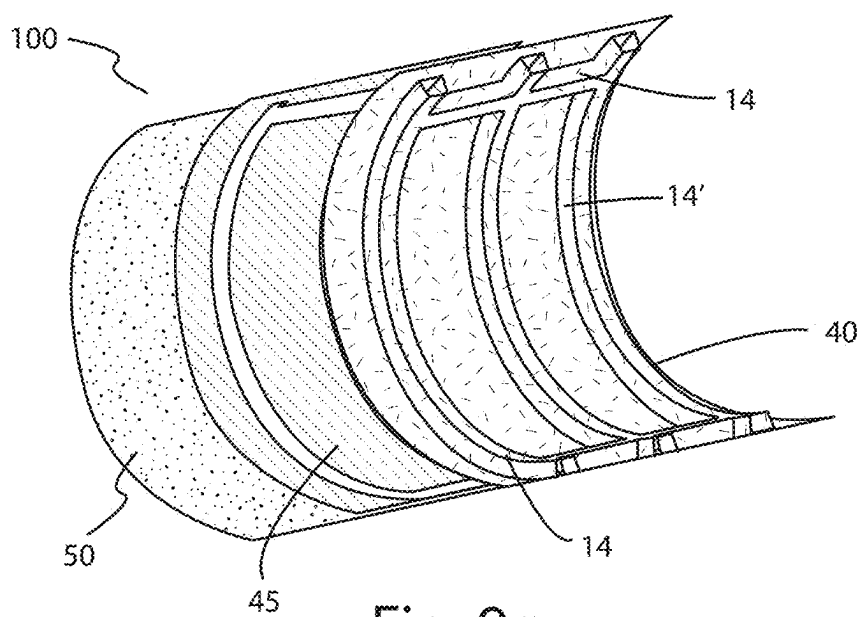


Fig. 9a

AIRCRAFT DOOR AND PRODUCTION METHOD

TECHNICAL FIELD

[0001] This invention relates to a door and in particular to a door that can be opened, made of composite material, for the aeronautical sector.

[0002] More in detail, the invention relates to a so-called Fan Cowl Door, that is an inspection hatch of a container (so-called gondola or nacelle) for the engine of an airliner, and its manufacturing method.

BACKGROUND

[0003] The inspection doors of the engines, which are made as an openable semi-cylindrical portion of the nacelle, are generally made of a composite material with a thermo-setting matrix with carbon fibre filler to ensure a low weight.

[0004] During assembly, oblong longitudinal and transverse reinforcements and elements for connecting the door to the nacelle are applied on the concave face of the door, which faces an internal compartment of the hatch, which are elements that protrude from the body of the door itself.

[0005] A disadvantage of the doors of known type is that the composite material used, which meets the structural requirements, is not resistant to the passing flame. In order to prevent degradation of the inner surface of the door in the event of a fire in the engine area, it is necessary to cover the surface itself with thermal protection panels that require frequent replacement and make it difficult to inspect critical internal areas.

[0006] With this thermal insulation mode, the connection areas between the door and the nacelle and the reinforcing elements remain in any case without protection.

[0007] A further disadvantage of the doors of known type is therefore that in the event of a fire in the engine, the door is at risk of detachment from the nacelle.

[0008] An escape of the flames from the nacelle would allow the flames to strike the areas of the wing and the fuselage which, being made, respectively, of aluminium alloy and carbon fiber, would completely lose their mechanical and structural characteristics.

SUMMARY

[0009] For this reason, the technical problem raised and resolved by the invention is that of providing a door for a nacelle and a method for making the door itself which allows the above-mentioned drawbacks of the prior art to be overcome.

[0010] This problem is solved by a door according to claim 1 and, according to the same inventive concept, by a method according to claim 9.

[0011] Preferred features of the invention are present in the dependent claims.

[0012] The invention provides some significant advantages.

[0013] The invention allows the manufacture of a multilayer door, configured to withstand high operating temperatures, thus preventing structural degradation in the event of a fire in the inner compartment of the nacelle.

[0014] A further advantage is that the multilayer door according to the invention comprises stiffening means formed in the main body of the door itself, allowing a structural uniformity of the construction material which, at

the high mechanical capacities, associates a contained weight and thickness and conductivity values which optimize its use in the aeronautical field.

[0015] A yet further advantage of the innovation is represented by the saving in weight, which amounts to about 5%, of the door with respect to the prior art doors.

[0016] The weight reduction combined with the characteristic of preventing the passage of flames in the event of an engine fire and of preserving the integrity of the aerodynamic surface of the nacelle, translates into the opportunity to maintain the aerodynamic performance of the aircraft as a whole in an operating condition.

[0017] A still further advantage is that the production method of the multilayer door according to the invention is quick and economical.

[0018] Other advantages, features and the means of use of the invention will become clear from the following detailed description of some embodiments, provided by way of example and without limiting the scope of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

[0019] Reference is made to the accompanying drawings, in which:

[0020] FIG. 1 is a perspective view of a door of known type;

[0021] FIGS. 2a-2c show a schematic view of the processing steps of an inner layer of an embodiment of the door according to the invention;

[0022] FIGS. 3a-3e show a schematic view of further processing steps for obtaining a first portion of an embodiment of the door according to the invention;

[0023] FIG. 4 shows an exploded perspective view of the first portion of an embodiment of the door according to the invention;

[0024] FIG. 5 shows a perspective view of a forming half-mold of the first portion of an embodiment of the door according to the invention;

[0025] FIG. 6 shows a perspective view of a forming counter-mold of a second portion of an embodiment of the door according to the invention;

[0026] FIG. 7 shows a perspective view of a mold for forming the multilayer door according to the invention;

[0027] FIG. 8 shows a perspective view of an embodiment of the door according to the invention;

[0028] FIG. 9a shows an exploded perspective of FIG. 8;

[0029] FIG. 9b shows an enlarged detail of FIG. 9a.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0030] The description below relates to a door which can be opened, in particular to a door of an engine gondola, or nacelle, configured to allow access to the internal compartment of the nacelle, in particular in the case of inspection and/or maintenance of the engine, and the method of manufacturing the door itself.

[0031] A first embodiment of the multilayer door according to the invention, as shown in FIG. 8, is denoted in its entirety with the numeral 100.

[0032] The door 100 according to the invention comprises a first multilayer portion 40, facing towards the inside of the nacelle in an assembly configuration, and a second layered portion 50, facing towards the outside of the nacelle in an assembly configuration.

[0033] The first portion **40** of the door **100** comprises at least three layers superimposed on each other in the direction of a thickness of the door itself.

[0034] In particular, the first portion **40** comprises an inner layer **10**, an outer layer **30**, and an intermediate layer **20** made of composite material, with an inorganic matrix.

[0035] Both the inner layer **10** and the outer layer **30** are made, for example in the form of a laminate, made of a composite material comprising a carbon fiber filler—preferably a carbon fiber fabric—and a thermosetting polymer matrix, preferably epoxy.

[0036] As regards the intermediate layer **20** of the door **100**, it is a thermal insulating layered element made of composite material, comprising an inorganic-based matrix and a carbon-based filler.

[0037] In particular, the inorganic-based matrix is a ceramic-based resin of the type developed by the same applicant and described in international patent application WO2018179019.

[0038] Advantageously, the ceramic-based resin used has a T_g (glass transition temperature) of 900°C . and can withstand a fire temperature of up to 1200°C ., in accordance with the provisions of ISO 2685.

[0039] The thermal stability of the ceramic-based resin developed by the applicant was tested over the operating temperature range and proved to be resistant.

[0040] For this reason, the intermediate layer **20** thus configured acts as an insulator, that is, as a thermal insulator, thanks to the porous structure of the ceramic-based matrix (which in particular has a percentage of residual porosity of between 16% and 21%) which is unalterable in a temperature range of between -55°C . and 310°C ., and complies with the aeronautical FTS regulations.

[0041] Advantageously, the intermediate layer **20** of the door according to the invention has a specific weight of between 1180 and 1280 kg/m^3 .

[0042] According to a first variant embodiment of the intermediate layer **20** the carbon fiber filler is in the form of a carbon fiber fabric, for example having a weight of about 200 g/m^2 .

[0043] This weight guarantees an optimal workability and impregnation of the carbon fiber fabric with the ceramic-based resin and allows a pre-impregnated fabric thickness of approximately 0.25 mm to be obtained.

[0044] Preferably, the intermediate layer **20**, for example made as a preform by overlapping four carbon fiber fabrics impregnated with a ceramic matrix (AS-HT), has a minimum thickness of at least 1 mm in such a way as to guarantee that the component cannot be perforated in case of contact with a flame with temperatures of 1200°C . and heat flux of 120 kW/m^2 .

[0045] According to a variant embodiment of the intermediate layer **20** the carbon fiber filler is in the form of a non-woven fabric, or felt.

[0046] In particular, a single layer of non-woven fabric is made, impregnated with a ceramic matrix (AS-HT), having a minimum thickness of about 2 mm, with a specific weight of about 600 kg/m^3 .

[0047] The 2 mm thickness of the non-woven fabric layer, impregnated with ceramic matrix (AS-HT), guarantees that the component cannot be perforated, for example in the event of contact with a flame with temperatures of 1200°C . and a heat flux of 120 kW/m^2 .

[0048] For this reason, the structural integrity of the intermediate layer **20** is guaranteed at the operating temperatures of the door in an operating phase, even in the event of flames in the engine compartment.

[0049] Advantageously, the non-woven fabric is obtained from the recycling of carbon fiber waste and therefore has a lower cost than a layer of carbon fiber fabric.

[0050] Moreover, the installation of a single layer of non-woven fabric requires less manpower during the construction of the door, with respect to the positioning of the layers of carbon fiber fabric, allowing a faster assembly of the door itself.

[0051] The door of a nacelle is an element shaped like a portion of a cylinder, being an integral part of the casing of the nacelle itself.

[0052] In order to ensure the structural rigidity and prevent bending or distortion during use, and during the opening and closing operations, the door is stiffened with longitudinal and transverse stiffening elements generally fixed to the internal surface of the door itself, as shown in FIG. 1.

[0053] Advantageously, the door described herein provides longitudinal and/or transversal reinforcement elements integrated in the body of the door itself.

[0054] In particular, the layers of the first portion **40** are shaped in such a way as to present at least one longitudinal protuberance and/or at least one transverse protuberance sized to increase the structural strength of the door **100**, acting as a structural stiffening element.

[0055] In this way, the formation of “flame proof” stiffening elements (Fire Proofness) is therefore guaranteed, in accordance with the provisions of the ISO 2685 standard.

[0056] According to the preferred embodiment, as shown in FIGS. 8, 9a and 9b, a first multilayer portion **40**, facing towards the inside of the nacelle in an assembly configuration, has at least one longitudinal protuberance and/or at least one transverse protuberance, which extends towards the inside of the nacelle.

[0057] In particular, each protuberance is obtained through a curvature of the layers that make up the first portion **40**. For this reason, at each protuberance on the face facing towards the inside of the nacelle, that is, on the exposed face of the inner layer **10**, there is a recess on the face of the first portion **40** facing towards the outside of the nacelle, that is, on the exposed face of the outer layer **30**.

[0058] For example, each protuberance has a U-shaped cross section. In particular, in order to optimise the structural stiffening, the cross section is substantially trapezoidal with two lateral segments inclined towards a lower base facing the inside of the nacelle.

[0059] The second portion **50** of the door **100** according to the invention is configured to be coupled to the outer layer **30** of the first portion **40**, to meet the aerodynamic requirements of the nacelle.

[0060] The second portion **50**, is made with an aerodynamic profile of composite material comprising a carbon fiber filler—preferably a carbon fiber fabric—and a thermoplastic polymer matrix (for example containing PMMA) which has a good resistance to impact. According to a variant embodiment, the second portion **50** has a thermosetting polymer matrix.

[0061] The invention also relates to an extremely simplified manufacturing process for the door.

[0062] The multilayer door is in fact composed of a core, or intermediate layer, made of carbon with a high porosity

inorganic matrix (AS-HT) which acts as a flame-resistant barrier and which is covered on each main face by at least one carbon fiber fabric and polymer matrix. Depending on the specific structural requirements, and on the dimensions of the door, on the number of attachment points with the fixed part of the nacelle and on the loads during a flight condition of the aircraft, carbon fiber fabrics can be used overlapping one another, in order to optimize the structural strength of the invention.

[0063] The door according to the invention also guarantees extreme simplicity of inspection, being manufactured in a single piece with a mainly constant thickness, without thermal protections which generally become soiled with engine fluids.

[0064] The manufacturing process of the door according to the invention envisages a first step of manufacturing a heat-insulating layered element **20**, that is to say, the intermediate layer, which in particular can be stored as a preform.

[0065] FIG. **2a** shows a half-mold (**S1**) with a substantially semi-cylindrical shape. In particular, the outer surface of the half-mold has at least two transverse grooves and/or at least two longitudinal grooves which appear as an equal number of transverse and longitudinal protuberances at the inner surface.

[0066] The intermediate layer **20** is then made by positioning the carbon filler, impregnated with the ceramic matrix (AS-HT), on the outer surface of the substantially semi-cylindrical half-mold.

[0067] In particular, the layers of carbon fiber fabric superimposed on each other and previously impregnated are arranged on the half-mold (**S1**).

[0068] According to a preferred variant embodiment, the carbon filler is in the form of a layer of non-woven fabric, in particular a non-woven fabric made of recycled carbon fiber material, previously impregnated.

[0069] The mold is then inserted into a vacuum bag and the material is then subjected to a first hardening process in an autoclave at a pressure of about 3 Atm, at a temperature of about 80° C. for a time of about 12 hours. The bag is then removed to facilitate the subsequent drying process.

[0070] The material is then subjected to a second heat treatment cycle, at ambient pressure, at a temperature of about 80° C. for a further period of about 12 hours.

[0071] Subsequently, the preform **20** made of carbon and ceramic matrix (AS-HT) is thermally treated at a temperature of about 750° C. in an inert gas atmosphere to achieve the desired chemical-physical characteristics. As shown in FIG. **2c**, the preform **20** is then separated from the half-mold (**S1**) and can be stored whilst waiting to be used in a subsequent lamination step, as shown for example in FIGS. **3a-3e**.

[0072] The lamination phase comprises the use of a second forming half-mold (**S2**), substantially semi-cylindrical, which differs from the half-mold (**S1**) only by the offset determined by the thickness of the layer **10**, on which is positioned an inner pre-impregnated layer **10**, on the inner pre-impregnated layer **10** is positioned the preform **20** and on the preform **20** is positioned an outer pre-impregnated layer **30**.

[0073] The assembly obtained is placed under a vacuum, in a special vacuum bag, to guarantee an adhesion of the inner pre-impregnated layer **10** and the outer pre-impregnated layer **30** on the inner and outer faces of the preform **20**

and in order to eliminate any air accumulated during application of the layers **10** and **30** on the pre-impregnated preform **20** (AS-HT).

[0074] In order to allow the polymerization of the matrix of the layers **10** and **30** and an optimal coupling of the same on the intermediate layer **20**, the above-mentioned assembly is inserted inside an autoclave, for a curing process, with a pressure value of 3 bar, at a temperature of 177° C., for a time of about 2.5 hours.

[0075] Advantageously, the method according to the invention allows at least one protuberance **14**, **14'**, or stiffening element to be made, during the forming of the layers of the first portion **40** of the door **100**, as shown in FIGS. **2a** to **4**. In addition to an advantage in terms of assembly and labour times and costs, the stiffening elements **14**, **14'** made of flame-resistant material, allow further improvement of the mechanical resistance of the door **100** in the event of a fire in the engine compartment.

[0076] FIG. **8** shows a preferred version of the door according to the invention comprising two transverse stiffening protuberances **14** and three longitudinal stiffening protuberances **14'**. However, according to the structural requirements required by the specific application, the door according to the invention can have a different number of stiffening elements, through the use of half-molds different from those shown in the drawings.

[0077] Preferably, the layers **10** and **30** have an excess resin content, which acts as an adhesive in the coupling phase between the layers.

[0078] In particular, a protective layer, called peel ply, that is to say, a polyester fabric which impregnates itself with the resin of the outer layer **30** but does not allow its coupling, is positioned between the outer layer **30** and the vacuum bag.

[0079] For this reason, peel ply is used to separate the first portion **40** from the vacuum bag and protect the surface of the outer layer **30** during the removal of the bag.

[0080] The vacuum bag is removed after the curing step and the first portion **40**, considered as a hybrid reinforcement portion of the door **100**, is left in the mold without removing the protective peel ply whilst awaiting subsequent gluing with the second portion **50**, that is, with the aerodynamic portion. In particular, the first portion **40** can be stored as a preform.

[0081] The second portion **50** is made by means of a further half-mold (**S3**), having a substantially semi-cylindrical shape.

[0082] As shown in FIG. **6**, a pre-impregnated layer of composite material comprising a carbon fiber filler and a thermoplastic matrix, in particular in polymethylmethacrylate (PMMA), is placed on the half-mold (**S3**) and subjected to a curing cycle in a vacuum bag, at ambient temperature for about 4 hours.

[0083] In order to proceed with the coupling, for example by lamination, of the second portion **50** on the first portion **40**, the protective layer is removed from the outer layer **30**. The surface on which the protective layer was applied is clean and ready for subsequent gluing or painting operations, avoiding abrasive cleaning operations.

[0084] As shown in FIG. **5**, once the protective layer has been removed, a layer of film **45** of epoxy structural adhesive is applied to the outer face of the layer **30**.

[0085] After positioning the adhesive film **45**, the two half-molds **S2** and **S3** are then assembled, as shown in FIG. **7**, by matching the second portion **50**, which acts as the

aerodynamic layer of the door **100**, with the first portion **40** which acts as a structural reinforcement, and the product is placed in a vacuum bag to proceed with the curing process of the adhesive in an autoclave, at a pressure of about 3 bar at a temperature of about 177° C. and for a time of about 2 and a half hours.

[0086] The first portion **40** of the door **100** is made in such a way that it does not have to be modified to meet the aesthetic specifications of the inside of the nacelle; therefore, once extracted from the forming mold, the inner layer **10** does not need to be worked or painted.

[0087] The multilayer door **100** according to the invention therefore appears as a carbon fiber laminate with excellent mechanical characteristics which, in the case in which the layers **10** and **20** are made of carbon fiber, with thermosetting matrix, are the following:

TABLE 1

ASTM	Value (MPa)
D3039 - Tensile Strength	365
D3039 - Tensile Modulus	55000
D695 - Compressive Strength	5.38
D695 - Compressive Modulus	51000
D6484 Open hole compression	125

[0088] With reference to what has been described above, the advantages of the invention are therefore lower weight of the door, increase in the general safety of the aircraft in the event of fire, easier maintenance, reduction of the components necessary for manufacturing the door and lower overall cost.

[0089] The invention allows the manufacture of a door substantially free of areas or components which are not fully protected, that is to say, isolated from the environment inside the nacelle, thus ensuring continuity and uniformity of the structural resistance along the entire surface extension of the door and a high resistance to perforation due to flames in case of fire.

[0090] The invention is described by way of example only, without limiting the scope of application, according to a preferred embodiment, but it shall be understood that the invention may be modified and/or adapted by experts in the field without thereby departing from the scope of the inventive concept, as defined in the claims herein.

1-10. (canceled)

11. An aircraft door, configured to be moved from a closed position of an outer shell of a nacelle to an open position to allow inspection of an internal compartment of the nacell, the aircraft door comprising:

- a first multilayer portion, comprising an inner layer, an outer layer and an intermediate layer, wherein said intermediate layer is made of composite material having an inorganic-based matrix and a carbon-based filler, said first portion shaped in such a way as to present at least one substantially oblong stiffening protuberance;
- a second layered portion having an aerodynamic profile and shaped to be coupled to said first multilayer portion,

the overall configuration of the door being such that, in an assembled configuration, and in a coupling between said first portion and said second portion, said at least

one substantially oblong stiffening protuberance is facing the internal compartment of the nacella.

12. An aircraft door according to claim **11**, wherein said carbon-based filler of said intermediate layer is a carbon fiber fabric.

13. An aircraft door according to claim **11**, wherein said carbon-based filler of said intermediate layer is a non-woven fabric, optionally of recycled carbon.

14. An aircraft door according to claim **11**, wherein said second portion is made of composite material having a thermoplastic or thermosetting polymer matrix and a carbon-based filler.

15. An aircraft door according to claim **11**, wherein said inner layer and said outer layer are made of composite material comprising a carbon fiber filler and a thermosetting polymer matrix.

16. An aircraft door according to claim **11**, wherein said first multilayer portion has a substantially oblong recess on said outer layer at each substantially oblong stiffening protuberance on said inner layer.

17. An aircraft door according to claim **11**, further comprising a layer of adhesive film material positioned between said first multilayer portion and said second portion.

18. An aircraft door according to claim **11**, comprising two transverse stiffening protuberances and three longitudinal stiffening protuberances.

19. A manufacturing method of an aircraft door, the aircraft door configured to be moved from a closed position of an outer shell of a nacelle to an open position to allow inspection of an internal compartment of the nacell, the aircraft door comprising:

- a first multilayer portion, comprising an inner layer, an outer layer and an intermediate layer, wherein said intermediate layer is made of composite material having an inorganic-based matrix and a carbon-based filler, said first portion being shaped to present at least one substantially oblong stiffening protuberance;
- a second layered portion having an aerodynamic profile and shaped to be coupled to said first multilayer portion,

the overall configuration of the door, being such that, in an assembled configuration, and in a coupling between said first portion and said second portion, said at least one substantially oblong stiffening protuberance is facing the internal compartment of the nacella, the manufacturing method comprising the steps of:

- (a) providing a first multilayer portion comprising a heat-insulating layered element, said first portion positioned on a forming half-mold;
- (b) providing a second portion having an aerodynamic profile, said second portion positioned on a further forming half-mold;
- (c) applying said second portion on said first portion by closing said further half-mold on said half-mold.

20. The manufacturing method of aircraft door according to claim **19**, wherein, before said step (c), there is a step of positioning a layer of adhesive film material on said first portion multilayer, the method further comprising a step (d) providing a step of curing said layer of adhesive film material.

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